Determination of the Neutrino Time of Flight Fermilab to Ash River Nova-DocDB-11860

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1 Executive Summary - NOνA Timing Verification

The timing of the NOνA detectors with respect to the NuMI beam spill was validated through multiple stages of calculations and experimental validation. The general stages of the verification procedure proceeded as:

1. A priori calculation of accelerator signal propagation
2. Measurement of the signal propagation to the Nova and Minos timing electronics
3. Observation of the neutrino rate excesses in the NOvA Surface detector
4. Comparison of the MINOS near detector timing to the NOvA near detector and accelerator timings
5. A priori calculation of the time of flight delays between the near and far NOνA detector locations
6. Measurement by MINOS of the observed delays between the crossing of the beam at Minos near detector and the Minos far detector, as measured by the Minos master clock located at the Soudan Lab
7. Measurement by MINOS of the phase difference between the Minos master cesium clock and the GPS 1 second boundary
8. Measurement via a time transfer procedure using a rubidium clock between the MINOS cesium master clock and six independent GPS receiver systems located at Ash River
9. Measurement of the timing delays between the GPS based NOvA master clock and the clocks signals that are propagated to the detector for synchronization and readout

In particular the time transfer between the Soudan lab and the Ash River lab utilized a high precision cesium based atomic clock which was located at Soudan Lab. The cesium atomic clock was used to train and phase align a portable rubidium based atomic clock. The training was performed over a period of approximately 24 hours. After the training of the rubidium clock, its phase was measured relative to the global positioning systems (GPS) one second boundaries signals at Soudan using a time interval counter with a precision for ΔT better than 100 ps. The phase measurement was determined to have a Gaussian error with $\sigma = 295ps$ as shown in Fig. 1.

The rubidium clock was transported by automobile to the Ash River lab. During this period the clock was powered by a large capacity uninterruptible power supply (battery backup) and monitored to ensure that clock was not interrupted during the trip. The duration of the transport was roughly 2.5 hours. Upon arrival at Ash River the rubidium clock was moved to the data acquisition computing facility where the NOvA timing systems are housed. The phase difference between the GPS 1 second boundary provided by the NOvA timing system deployed at Ash River was then measured using the same time interval counter that was used at Soudan lab. These phase intervals were measured for each of the three GPS receivers which had been in operations at the lab. The time interval between the rubidium clock's reference pulses and the NOvA timing system's master clock was then measured utilizing the NOvA timing system's internal diagnostics.

Three additional NOvA timing units were then deployed to Ash River lab and phases between these upgraded systems and the rubidium clock were measured in the same manners as the original three units. Tests were then performed of the the power on and initialization characteristics of the timing system to determine if the timing phases were stable across power cycling, synchronization and re-initialization of the systems.

The rubidium clock was then transported back to Soudan Lab and the phase differences between the rubidium clock and the MINOS cesium master clock were remeasured to determine the total drift in the rubidium clock over the time during which the measurements were performed. The clock was found to have a roughly linear drift of 0.45 ps/s accounting for a total accumulated drift of 16.5 ns

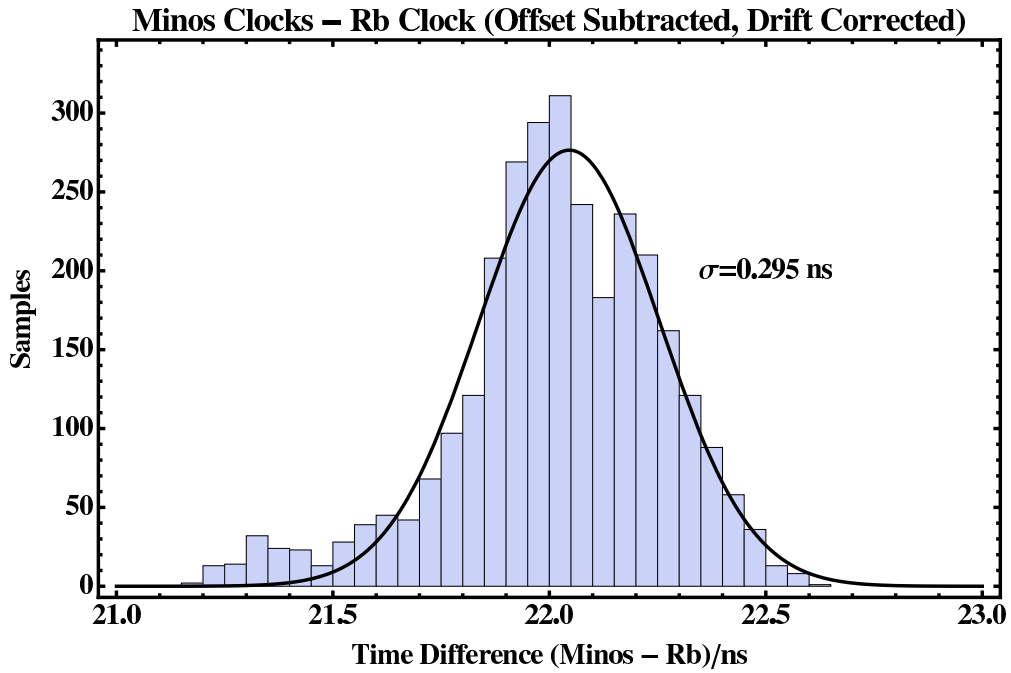


Figure 1: Minos master cesium clock to trained rubidium clock phase offset after drift corrections. Distribution is fit to a Gaussian profile for error determination.

1.1 Summary

The NOvA far detector timing system was determined to be correctly aligned to the GPS time standard. The six independent timing distribution systems which were measured at Ash River were found to be aligned to the GPS clock to better than 16 ns per unit as shown in Fig. 2. Measurement errors and systematics related to the stability of the measurements across timing system restarts are included.

The result of the measurements were that the time of the crossing of the neutrino beam at the front face of the Ash River far detector was predicted to occur 2918.06 μs after the reception of the MIBS \$74 accelerator signal by the NOvA master timing system located at in the Minos service building and extrapolated to the far detector by the NOvA timing and synchronization system.

Neutrino candidates were observed in the far detector at a timing offset corresponding to 2918.5 μs after the reception of the NuMI trigger t_0 time.

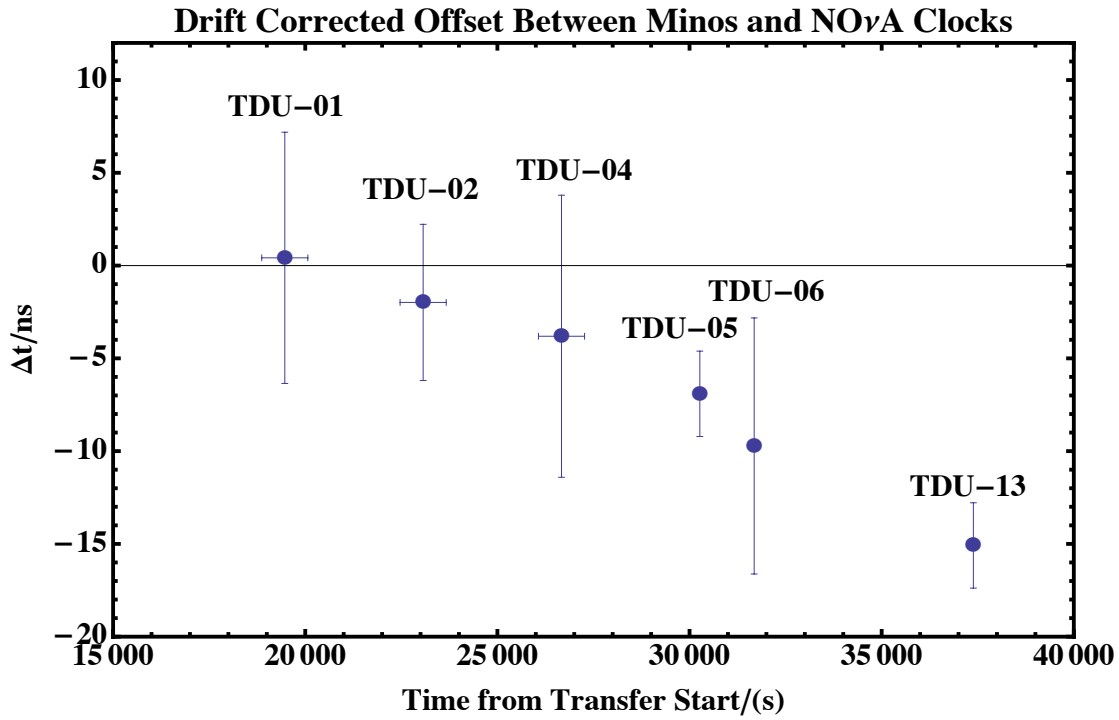


Figure 2: Alignment of the NOvA timing distribution systems (TDUs) to the MINOS clock systems at Soudan Lab. The measurements were corrected for drift of the rubidium atomic clock used for the time transfer and for known delays in the Soudan and Ash River timing systems. The time deltas are reported as a function of the number of seconds from the start of the first measurement at Soudan Lab.

2 Introduction

The operation of the NOvA data acquistion (DAQ) and trigger hardware/software requires knowledge of the flight path of the neutrinos that are generated in the NuMI (Neutrinos at the Main Injector) and travel to the far detector. Knowledge of this flight path allows for the computation of the propagation time required for the neutrinos to travel from the their origin to their crossing point with the NOvA far detector. This propagation time is used to then set the global time delay that is used in the trigger system that determines the absolute start point of time window for which a "beam" trigger is issues.

The NuMI spill duration is nominally 10 μs and in baseline operations the trigger window would be set to be centered on this beam spill and include small side bands which allow for characterization of the detector in the interval surrounding the beam spill. In the NOvA TDR these sidebands were proposed to be ± 10 μs which would yeild a total trigger window of 30 μs. This trigger width is not optimal for the manner in which the DAQ systems were designed. The design of the DAQ readout systems relys on a natural time based discretization of the data. This discretization provides a smallest quanta of data which groups the hits into 50 μs long blocks. As a result, the event builders are unable to break the data into windows shorter than 50 μs and requests for the extraction of data windows with a given width always result in the actual extraction of an integer number of contiguous 50 μs blocks. As a result of this design the optimal setting for the trigger window is to center the 10 mus trigger inside of a 50 μs base window which then provides sidebands with a minimum width of 20 μs. The trigger window then has the form of a half open interval:

$$\Delta T_{trig} = [t_0 + t'_{delay} - 20\ \mu s, t_0 + t'_{delay} + 30\ \mu s) \tag{1}$$

Where t_0 is the time of the leading edge of the spill and t'_{delay} is the offset/delay between when the spill occured at FNAL and the time at which the beam crossed the detector.

When this trigger window is actually extracted from the data stream, it will be superimposed upon actual boundaries provided by the microslice quanta. This will in all cases cause it [the trigger] to overlap with exactly 2 microslice blocks. This is because the microslice intervals are considered closed:

$$\mu Slice = [t_n, t_n + 50\mu s] \tag{2}$$

So that in the event that the front edge of a trigger window exactly coincides with one of the microslice boundaries, the trigger window is considered in overlap with both the trailing edge of first microslice and the leading edge of the second microslice. This will cause both to be recorded into the data stream and for the final event on disk to have a width of 100 μs reguardless of the position of the trigger t_0 realative to the microslice boundary times.

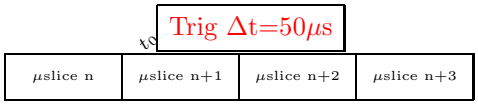


Figure 3: Event extracted from Microslice

In this scheme, the time t'_{delay} must be initially determined from an apriori calculation of the flight time of the neutrinos from the NuMI target to the NOvA detectors. This document details how this calculation was performed and how certain inputs, such as the position of the NOvA detector were determines.

3 Detector Positioning

The determination of the position of the far detector was determined using the information provided by the GPS receivers installed in the timing system masters (TDU masters) at Ash River. These three receivers

were located in the far detector computing center in relay rack 3 and connected to a set of three antennas that are mounted on the roof of the Ash River building.

The position of the near detector on the surface (NDOS) was determined through the use of the master TDU mounted in the first electronics rack at the NOvA surface building upper catwalk and connected to the antenna mounted on the roof of the building. Additionally the position of the MINOS surface building was determined in a similar manner using the TDUs mounted in the electronics racks in the MINOS surface building and connected to two separate antennas mounted on a mast on the roof of the building.

3.1 GPS Antenna Position

The Antennas on the Ash River building are mounted on a single mast with four mount points for Antennas. Each Antenna is mounted on a mount point at a distance XXX from the central axis of the mast. The antennas are then connected to the recievers through an approximately 100 foot cable run that is common to all the antennas and receivers.

At the MINOS service building a similar antenna installations is used with a single mast with two mount points and common cable runs down to the GPS receivers. For the NOvA surface building a single mast and antenna were mounted and only a single cable runs down from the antenna to the receiver.

3.2 TDU NMEA Data Stream

To extract the calculated GPS position, the receiver's NMEA data stream output was connected to a serial console using a USB to serial adapter (USB/serial chip on TDU). The console was brought up using the GNU "screen" utility using a connection speed of 38400 baud.

When this was started the MNEA data stream was continously sent to the serial console in a fashion where it repleats its NMEA block 1 once every 5 seconds. When extracted the blocks read similar to:

```
$GPGLL,4822.716417,N,09249.877638,W,210341.002,A,A*41
$GPRMC,210341.002,A,4822.716417,N,09249.877638,W,0.000,0.00,070313,,,A*40
$GPGGA,210341.002,4822.716417,N,09249.877638,W,1,12,0.71,382.823,M,-32.462,M,,*5D
$GPGSA,A,3,31,11,14,25,01,20,23,30,22,16,12,32,1.20,0.71,0.96*0C
$GPGSV,3,1,12,31,88,142,48,11,02,241,38,14,35,092,45,25,31,053,44*75
$GPGSV,3,2,12,01,20,256,44,20,40,300,45,23,11,285,42,30,27,171,44*74
$GPGSV,3,3,12,22,07,152,42,16,02,182,37,12,07,029,40,32,64,275,47*7C
$POLYT,210341.002,070313,421421.0022814,1730,421437.0022814,-281415,333.984,25,1127648768,00017,00017
$POLYP,210341.002,4822.716417,N,09249.877638,W,350.361,G3,001.9,005.1,0.000,0.00,0.000,,0.71,0.96,1.2
$POLYS,12,31,U,142,88,48,255,11,U,241,02,38,255,14,U,092,35,45,255,25,U,053,31,44,255,01,U,256,20,44,
$POLYI,JN,18,EXT,HPOS,VPOS,INT,TSYNC,*19
```

The $GPGGA string is then used to determine the position of the fix. In the case of the data above the $GPGGA string is decode to the position of the Ash River antenna array as shown in Table 1.

The $POLYP string can be similarly decoded to provide an alternative position fix based on a user datum ellipsoid and the $POLYT string provides the GPS time information along with the errors in the measurement and associated drifts.

4 Flight Path Determination

The flight path between the near detector and the far detector was computing in two different manners using different assumptions for the shape of the Eath. These calculations are done both for self consistancy and to determine the potential systemmatic error that is introduced by the model of the Earth being used.

Name	Data	Description
Sentence Identifier	$GPGGA	Global Positioning System Fix Data
Time	210341.002	21:03:41.002 Z
Latitude	4822.716417,N	48d 22.716417' N
Longitude	09249.877638,W	92d 49.877638' W
Fix Quality	1	0=Invalid, 1=GPS fix, 2 = DGPS fix
Number of Satellites	12	12 satellites in view and locked
Horizontal Dilution of Precision (HDOP)	0.71	final std error is 0.71 times input error
Altitude	382.823,M	382.823 meters above mean sea level
Height of geoid above WGS84 ellipsoid	-32.462,M	-32.462 meters
Checksum	5D	Checksum for sentence

Table 1: Decoding of the $GPGGA sentence for the Ash River GPS antenna array.

Location	Latitude	Longitude	Altitude (m)	Offset Depth to Detector (m)
Ash River Ant-1	48 d 22.715780"	92 d 49.876930	348	12+7.8
Ash River Ant-2	48 d 22.715780"	92 d 49.876930	348.638	12+7.8
Ash River Ant-3	48 d 22.716416"	92 d 49.877639	350.346	12+7.8
Fnal Minos	41 d 50 m 23 s	88 d 16 m 13 s	228	105 + 15

Table 2: GPS coordinates of NOvA detectors used for flight path calculations.

4.1 Spherical Earth

In the simplest approximation the Earth can be taken as a sphere with a mean equitoral radius $R_E = 6,378,137$ meters. In this representation the longitude (θ), latitude (ϕ) and altitude (A) of a point can be converted to a cartisan coordoinates in an Earth centered reference frame representation.

The transformation $\{\theta, \phi, A\}$ to $\{p_x, p_y, p_z\}$ is given by:

$$p_x = (R_E + A)\sin(\theta)\cos(\phi) \tag{3}$$
$$p_y = (R_E + A)\cos(\theta)\cos(\phi) \tag{4}$$
$$p_z = (R_E + A)\sin(\phi) \tag{5}$$

In this case the longitude and latitude are taken from the GPS NMEA data stream and the altitude that is used is the altitude above mean sea level.

In the case of the near detector to far detector the parameters sets that were used were:

Because the values returned by the GPS receivers are the positions of the antenna systems, offsets are applied to these reported positions to account for the difference between the location of the antenna and the location of the detector. In each case the offset is computed to correspond to the front face of the detector (face closest to the beam) so that the flight time delays that are computed are for the earliest arrival of particles between sites (i.e. delays are computing from front face crossing to front face crossing and from that corrected for position). The depth offsets are computed to the mid-point height of the detector (7.8 m for far det) and include the approximate height of the detector over the top edge (12 m for far det). For the NuMI cavern the depth of the cavern is taken to be 105 m and the mount point of the antenna on the Minos service building is taken as 15 m above the surface.

To comput these offsets the plane tangent to an earth centered sphere, at the GPS measured antenna position is taken, and the vector between antenna position and the center of the front face of the detector is calculated. The projects of the offset vector into the earth centered fixed reference frame are extracted and

Location	Δx_{offset} (m)	$Deltay_{offset}$ (m)	$Deltaz_{offset}$ (m)	Notes
Ash River Ant Base	TBD	TBD	TBD	mount axis
Ash River Ant-1	TBD	TBD	TBD	south ant.
Ash River Ant-2	TBD	TBD	TBD	north ant.
Ash River Ant-3	TBD	TBD	TBD	west ant.
Fnal NDOS	2	3	-3	105 + 15

Table 3: Postions offsets used to correct for antenna verse detector front face positions.

the earth centered fixed coordinates of the antenna are shifted accordingly.

The standard Euclidian metric is computed between the ECF coordinates of the the near and far detectors to determine the straight line flight path between the detectors.

$$\Delta L = \sqrt{\sum_i ((p_i - \delta p_i) - (p'_i - \Delta p_i^{prime})^2} \tag{6}$$

$$= 810,319 \text{ m} \tag{7}$$

The neutrinos are assumes to have a speed equal to the speed of light, c. The resulting flight time is then:

$$\Delta t = \Delta L / c \tag{8}$$

$$= 2703.3 \ \mu\text{t} \tag{9}$$

4.2 WGS84 Earth Model

Due to the lattitudes and distances between the near and far detector sites, a more accurate determination of the flight path can found by using a deformed Earth model. The standard description for the deformation can be taken from the WGS84 model which establishes elliptical major and minor axis for the earth and a set of transformations from the ellipsoid into an Earth Centered Fixed (ECF) representation.

The parameters used for the WGS84 model, as used by the both the NOvA and Minos experiments are given by:

$$R_a = 6378137 \text{ m} \tag{10}$$

$$R_b = 6356752.31424518 \text{ m} \tag{11}$$

$$\epsilon_a = \sqrt{\frac{R_a^2 - R_b^2}{R_a}} \tag{12}$$

$$\epsilon_b = \sqrt{\frac{R_a^2 - R_b^2}{R_a}} \tag{13}$$

With a prescription for transforming from the GPS reported (θ, ϕ, A) coordinates into the ECF coordinates (p_x,p_y,p_z).

$$R_{curv}(\theta) = \frac{a}{\sqrt{1-\epsilon_a^2 \sin\theta^2}} \tag{14}$$

$$p_x = (R_{curv}(\theta) + A)\cos\theta\cos\phi \tag{15}$$

$$p_y = (R_{curv}(\theta) + A)\cos\theta\sin\phi \tag{16}$$

$$p_z = R_b^2/R_a^2(R_{curv}(\theta) + A)\sin\theta \tag{17}$$

In the case of the path length between the FNAL and Ash River timing stations the path length is taken by comparing the distance between the ECF coordinates of far detector and the Minos service building position (offset to the beam depth and detector cavern).

$$\Delta L = \sqrt{\sum_i ((p_i - \delta p_i) - (p_i' - \Delta p_i^{prime})^2} \tag{18}$$

$$= 809,606 \text{ m} \tag{19}$$

The time of flight is then computed assuming the propagation of the beam at the speed of light

$$\Delta t = \Delta L/c \tag{20}$$

$$= 2700.56 \ \mu\text{s} \tag{21}$$

This time offsite is the delay between when the beam crosses the near detector cavern and when the beam should cross the front face of the far detector.

References